

Hilal Morrar

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EDUCATION

University of Texas at Austin

Masters of Science, Computer Science

University of California, Santa Cruz

Bachelors of Science, Computer Science

August 2023 – May 2025

Austin, Texas

September 2018 – June 2022

Santa Cruz, CA

EXPERIENCE

Software Engineer Intern

May 2024 – August 2024

Cisco Systems - Duo Security

San Francisco, CA

- Engineered a new feature to an internal performance load testing tool using Python, AWS EC2, and MySQL to dynamically create simulated customers and production HTTP requests.
- Created reports for software release tests executed with GitHub CI/CD pipelines to monitor performance metrics for **50,000+ concurrent users** with Datadog, Grafana, and Kibana dashboards for observability.
- Resolved faulty pod monitoring in Kubernetes deployments using Python to eliminate silent job failures or stalled pods and **saved 10% on AWS resources**.

Software Engineer Intern

May 2023 – August 2023

Stealth Startup

Remote

- Designed a RESTful API and implemented DELETE request functionality for a distributed, fault-tolerant, and consistent key-value store using Go and Docker.
- Developed internal API routes to support dynamic server coordination, data rebalancing, and Bash scripts to automate setup/teardown processes to **improve development efficiency by 30%**.
- Utilized Goroutines for concurrency when handling client and internal HTTP requests for system communication.

Machine Learning Engineer

September 2020 – August 2021

Applied Machine Learning Lab

Santa Cruz, CA

- Led the design and training efforts of various CNN model architectures for time-series data with PyTorch.
- Achieved **95% accuracy** by tuning machine learning models with hyperparameter optimization.
- Distributed model training with PyTorch DDP in a Kubernetes GPU cloud for a **50% reduction in training time**.

Frontend Software Engineer

July 2020 – September 2020

Resilient Renewable Efficient Energy Systems

Santa Cruz, CA

- Built a registration UI using React, JavaScript, and AWS S3 for an IoT device with hardware and product teams.
- Coordinated end to end integration with backend services on AWS EC2 to ensure smooth device onboarding.
- Optimized data handling and client-side input validation to **raise user experience and accessibility scores by 17%** based on Google Lighthouse audits.

PROJECTS

Bookstore Management Service | *GitHub, Go, Gin, PostgreSQL*

- Developed a RESTful HTTP API in Go with CRUD functionality for a bookstore to manage books, orders, and users.
- Integrated PostgreSQL with migrations for data storage and secured API routes with JWT authentication cookies.
- Facilitated consistent builds and workflows with Docker and GitHub CI/CD to containerize, test, and deploy.

JuiceFS | *AWS EC2, AWS S3, GitHub, Go, PostgreSQL, Python*

- Extended an open source, distributed file system to support a new object storage API to enhance performance in large-scale distributed environments.
- Implemented a storage interface in Go that resulted in **up to 14% improvement in read/write metrics** for larger files using AWS and PostgreSQL.
- Automated performance tests in Python with a dual benchmarking workflow to collect data and validate the custom storage backend.

TECHNICAL SKILLS

Languages: Assembly, C, C++, Go, HTML/CSS, Java, JavaScript, TypeScript, Python

Frameworks/Libraries: Agile, Gin, Keras, NumPy, PyTorch, React, TensorFlow, pandas, scikit-learn

Developer Tools: CI/CD, Docker, Git, Jira, Kubernetes, Linux, MySQL, PostgreSQL, OpenAPI, Slack, Swagger, Terraform